

A Temporal School Division Level Investigation of the Influence of Income, Exam Subject, and Exam Grade Level on Virginia Standards of Learning (SOL) Pass Rates

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DATASETS



Raw SOL Pass Rate Dataset



Raw Median Household Income Dataset



Geospatial dataset of VA school divisions
(kept separate from main dataset)

```
# view  
data_joined_final  
✓ 0.0s
```

MAIN DATASET

	school_year	division_name	subject	grade_level	pass_rate	med_income_year	med_income
0	2014-2015	accomack county	english:reading	3	61.08	2014	39389.0
1	2014-2015	accomack county	mathematics	3	64.65	2014	39389.0
2	2014-2015	accomack county	english:reading	4	66.57	2014	39389.0
3	2014-2015	accomack county	mathematics	4	76.16	2014	39389.0
4	2014-2015	accomack county	english:reading	5	70.45	2014	39389.0
...	...	...	...	...	...	...	...
7890	2018-2019	williamsburg-james city county	mathematics	6	80.95	2018	83048.0
7891	2018-2019	williamsburg-james city county	english:reading	7	83.28	2018	83048.0
7892	2018-2019	williamsburg-james city county	mathematics	7	86.58	2018	83048.0
7893	2018-2019	williamsburg-james city county	english:reading	8	77.96	2018	83048.0
7894	2018-2019	williamsburg-james city county	mathematics	8	NaN	2018	83048.0

7895 rows × 7 columns

A few example rows of the **main dataset** consisting of information on both SOL pass rates and median household incomes for school divisions between the 2014-2015 to 2018-2019 academic school years.

Variable Name	NOIR Type	Description
school_year	Ordinal	The academic school year, like 2014-2016. Note, this is considered ordinal because the order is meaningful, but the distance between each school year is not due to differing calendars.
division_name	Nominal	The name of the school division.
subject	Nominal	The subject of the SOL test.
grade_level	Ordinal	The grade level associated with the test.
pass_rate	Ratio	The pass rate as a percentage (%). A passing test score is 400-600 [1]
med_income_year	Interval	The year the median household income measurement is associated and calculated for.
med_income	Ratio	The median household income in 2023 dollar equivalent amount.

The variable names, NOIR types, and descriptions of the variables in the **main dataset**.

RESEARCH QUESTIONS



1. What are the differences in pass rates between SOL exams of subjects English:Reading and Mathematics and grade levels 3 to 8 between the 2014-2015 to 2018-2019 school years?
2. What are the temporal trends in pass rates between the 2014-2015 to 2018-2019 school years for SOL exams of subjects English:Reading and Mathematics and grade levels 3 to 8?
3. How are median household incomes trending across Virginia school divisions from 2014 to 2018?
4. Are SOL exam pass rates significantly influenced by median household income, the exam's subject (English:Reading and Mathematics), or the exam's grade level (3 to 8) between the 2014-2015 to 2018-2019 school years?
5. During the 2018-2019 school year, the school year we have the latest data available for, what Virginia school divisions have the highest and lowest median (typical) pass rates on the English:Reading and Mathematics SOL exams as well as the highest and lowest median (typical) household incomes?



METHODS

Raw Datasets:



SOL Pass Rate Dataset



Median Household
Income Dataset



Geospatial dataset of VA
school divisions

Data Cleaning:



Main
Dataset

Geospatial
Dataset

Data Analysis:



Traditional Visualizations, Summary
Statistics, and Regression Analysis



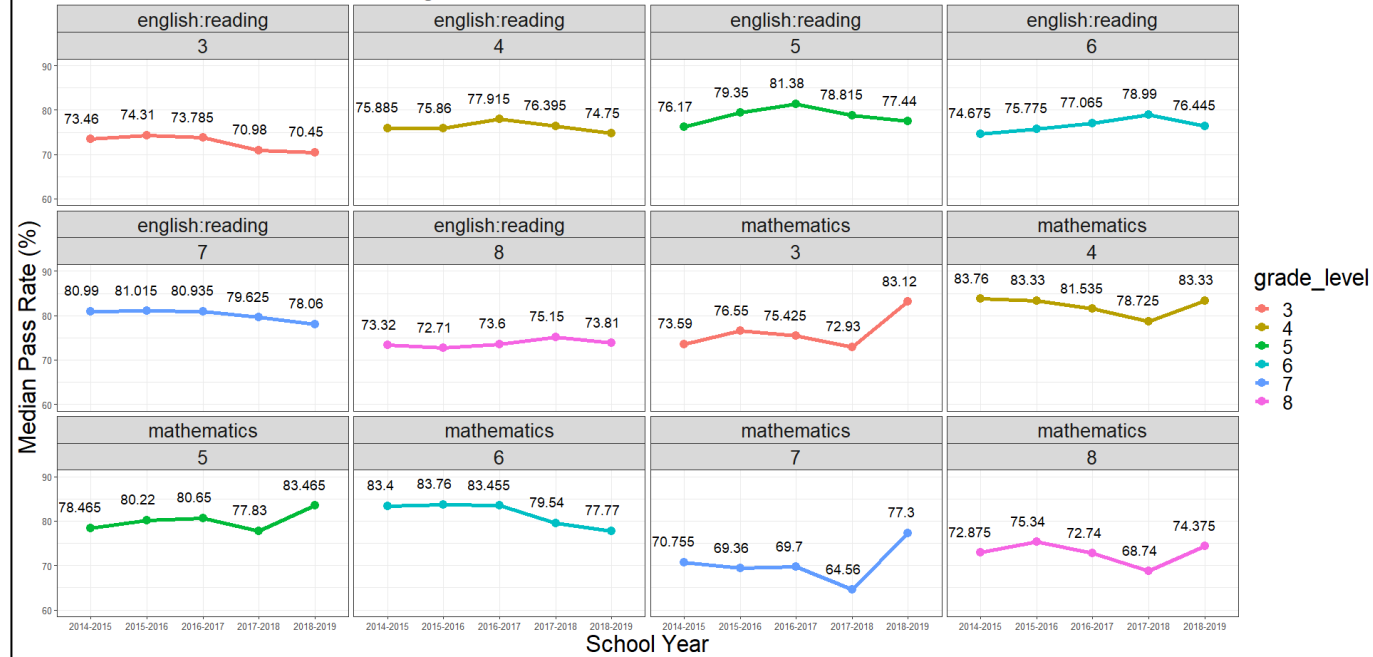
Summary Statistics



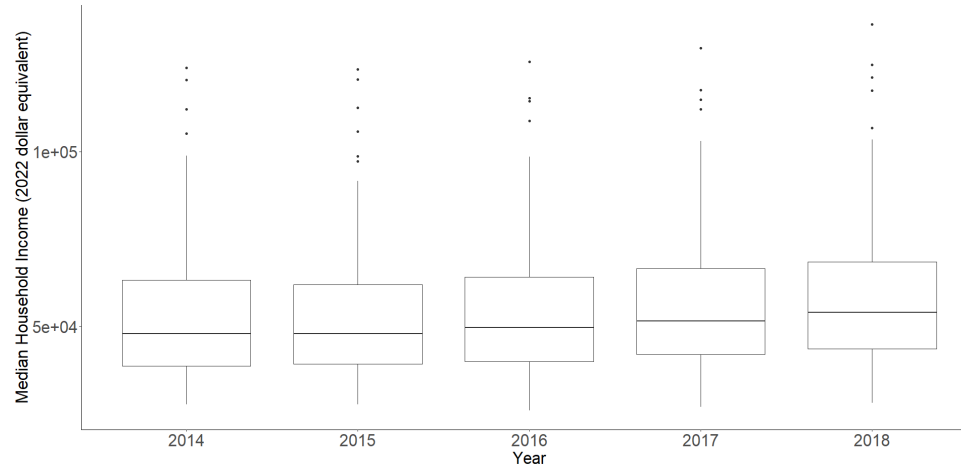
Choropleth Maps

Key Findings Part 1

Median Pass Rates (%) for Subjects English:Reading and Mathematics and Grade Levels 3 to 8 SOL Exams Across Virginia School Divisions over the 2014-2015 to 2018-2019 School Years



Distributions of Median Household Income (2022 dollar equivalent) across Virginia School Divisions from 2014 to 2018

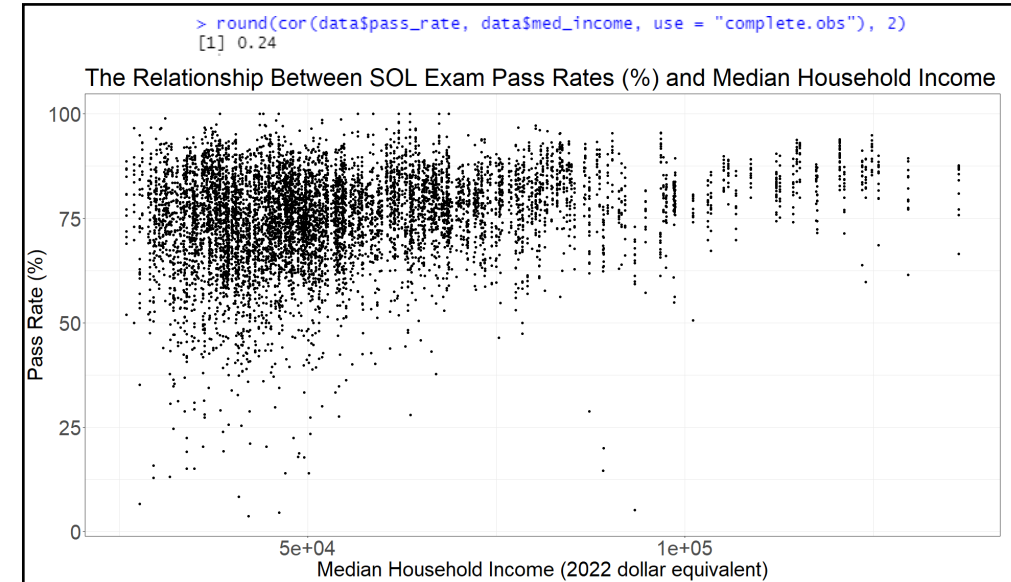
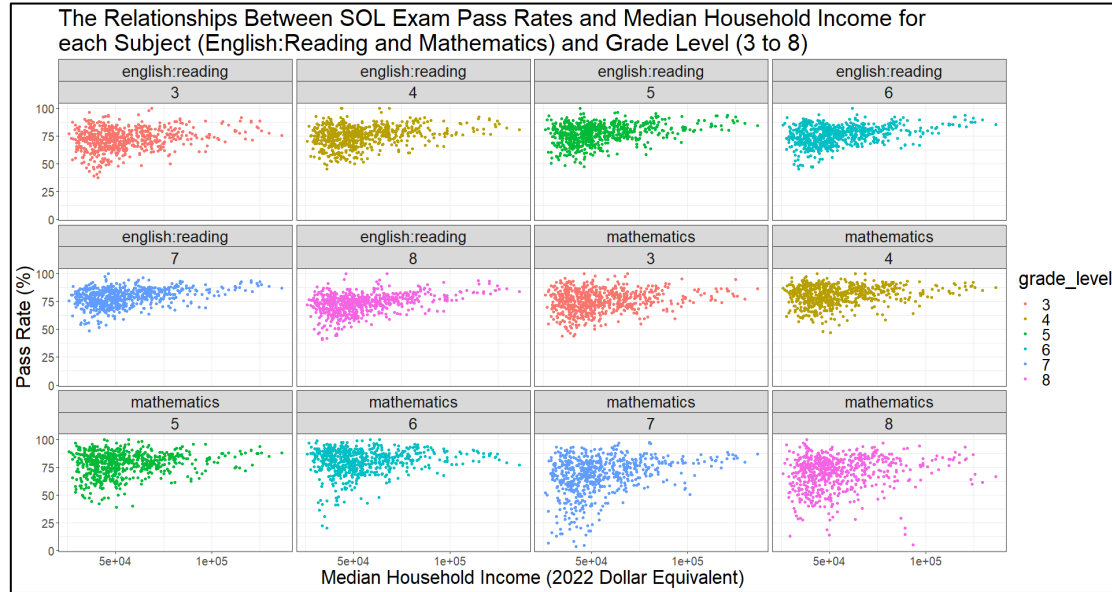


```
# A tibble: 5 x 5
# Groups:   school_year [5]
  school_year med_income_year median_med_income IQR skewness
<fct>         <int>         <dbl> <dbl> <dbl>
1 2014-2015      2014         47804 24350 1.40
2 2015-2016      2015         47911 22633 1.39
3 2016-2017      2016         49712 24195 1.32
4 2017-2018      2017         51495 24897. 1.31
5 2018-2019      2018         54197 24928 1.31
```

school_year	med_income_year	MIN(med_income)	AVG(med_income)	MAX(med_income)	STD(med_income)
2014-2015	2014	27746	52957	123966	19095.11
2015-2016	2015	27731	52959.84	123453	19058.3
2016-2017	2016	26000	54041.19	125672	19112.18
2017-2018	2017	26971	56216.43	129588	19433.87
2018-2019	2018	28071	58402.21	136268	20508.77



Key Findings Part 2



```
> # remove NA rows
> data_no_na = na.omit(data)
> model = lm(pass_rate ~ med_income + subject + grade_level + subject:grade_level, data = data_no_na)
> summary(model)
```

Call:
lm(formula = pass_rate ~ med_income + subject + grade_level + subject:grade_level, data = data_no_na)

Residuals:

Min	1Q	Median	3Q	Max
-69.943	-5.498	0.897	6.670	32.428

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	6.455e+01	5.149e-01	125.351	< 2e-16 ***
med_income	1.364e-04	5.908e-06	23.084	< 2e-16 ***
subjectmathematics	3.502e+00	5.655e-01	6.193	6.21e-10 ***
grade_level4	3.045e+00	5.655e-01	5.386	7.43e-08 ***
grade_level5	5.262e+00	5.653e-01	9.309	< 2e-16 ***
grade_level6	3.589e+00	5.653e-01	6.350	2.27e-10 ***
grade_level7	7.001e+00	5.650e-01	12.390	< 2e-16 ***
grade_level8	8.917e-01	5.650e-01	1.578	0.114580
subjectmathematics:grade_level4	2.463e+00	7.997e-01	3.080	0.002076 **
subjectmathematics:grade_level5	-2.648e+00	7.994e-01	-3.313	0.000928 ***
subjectmathematics:grade_level6	8.366e-01	7.992e-01	1.047	0.295232
subjectmathematics:grade_level7	-1.504e+01	8.028e-01	-18.737	< 2e-16 ***
subjectmathematics:grade_level8	-6.603e+00	8.035e-01	-8.218	2.40e-16 ***

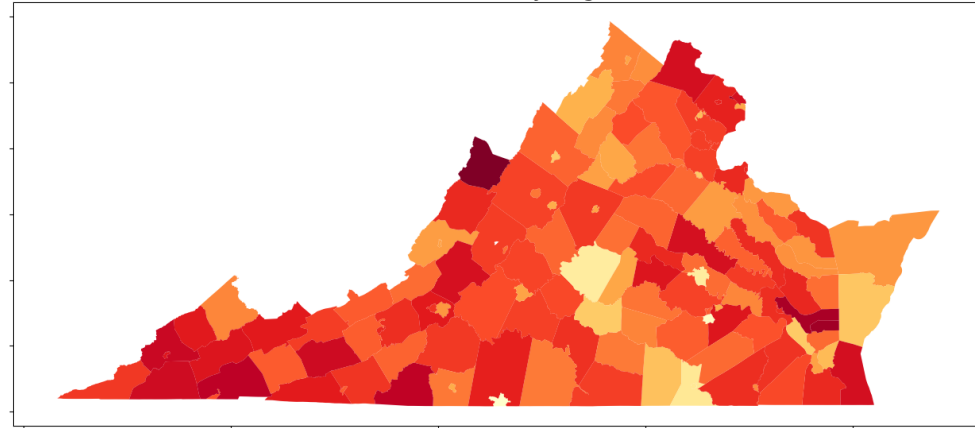
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.26 on 7844 degrees of freedom
Multiple R-squared: 0.1768, Adjusted R-squared: 0.1756
F-statistic: 140.4 on 12 and 7844 DF, p-value: < 2.2e-16



Key Findings Part 3

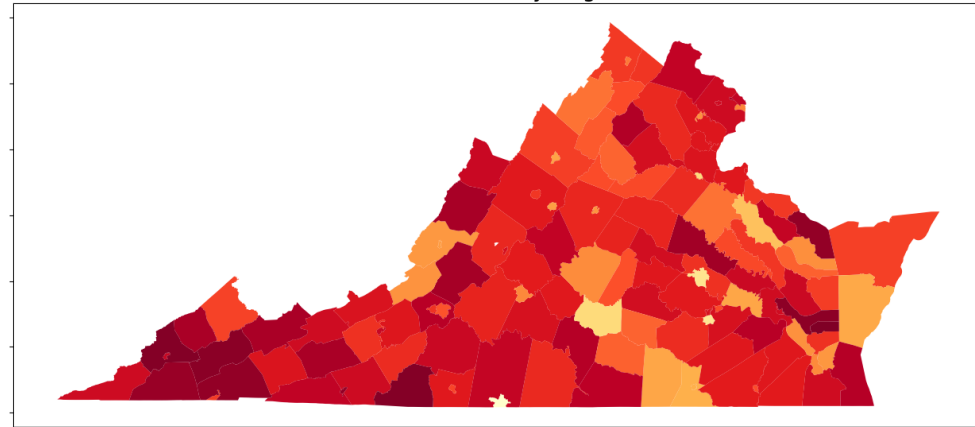
Median Pass Rates for English:Reading SOL Exams across Grade Levels 3 to 8
in the 2018-2019 School Year by Virginia School Division



Median Pass Rate

50 55 60 65 70 75 80 85 90

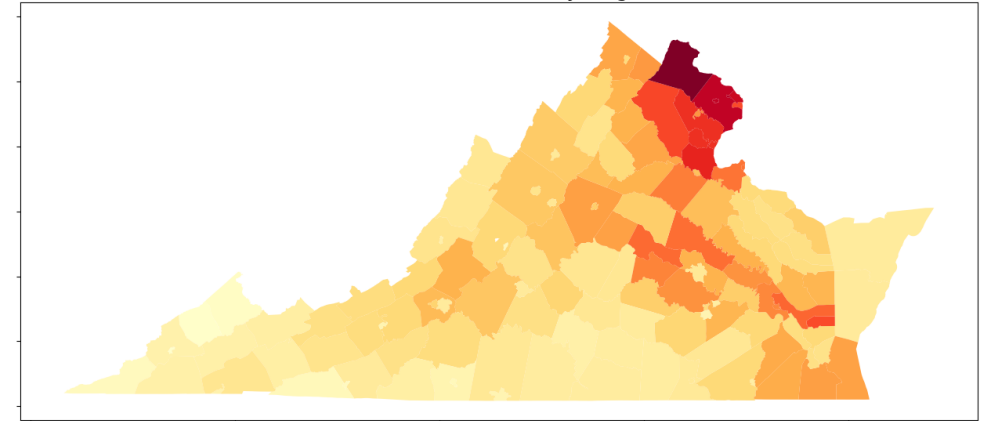
Median Pass Rates for Mathematics SOL Exams across Grade Levels 3 to 8
in the 2018-2019 School Year by Virginia School Division



Median Pass Rate

50 60 70 80 90

Median Household Incomes in 2018 by Virginia School Division



Median Household Income

40000 60000 80000 100000 120000



Lessons Learned

- I got practice coding, processing data, visualizing data, and computing summary statistics in Python, R, and SQL
- I practiced making choropleth maps in Python
- I learned how to manage my time on a difficult project



Images Used

- [question mark | 3d human with a red question mark | Damián Navas | Flickr](#)
- [https://vdoe.prod.govaccess.org/](#)
- [https://www.census.gov/](#)
- [File:Python-logo-notext.svg – Wikipedia](#)
- [https://en.wikipedia.org/wiki/R_%28programming_language%29](#)
- [Mysql workbench - Social media & Logos Icons \(icon-icons.com\)](#)

